

$$f(x) = \frac{x^3-x}{x^2-9} = x + \frac{8x}{x^2-9}$$

$$D_f = \mathbb{R} \setminus \{-3; 3\}$$

$$\text{Zeros: } f(x)=0 \Leftrightarrow x=0 \vee x=1 \vee x=-1$$

$$\lim_{x \rightarrow \pm\infty} f(x) = \lim_{x \rightarrow \pm\infty} \frac{x^3-x}{x^2-9} = \frac{\infty}{\infty} = \lim_{x \rightarrow \pm\infty} \frac{x^2 \left(x - \frac{1}{x} \right)}{x^2 \left(1 - \frac{9}{x^2} \right)} = \lim_{x \rightarrow \pm\infty} \frac{x - \frac{1}{x}}{1 - \frac{9}{x^2}} = \frac{\infty}{1} = \infty$$

$$f'(x) = \left(\frac{x^3-x}{x^2-9} \right)' = \frac{(x^3-x)'(x^2-9) - (x^2-9)'(x^3-x)}{(x^2-9)^2} = \frac{(3x^2-1)(x^2-9) - 2x(x^3-x)}{(x^2-9)^2} = \frac{3x^4-28x^2+9-2x^4+2x^2}{(x^2-9)^2} = \frac{x^4-26x^2+9}{(x^2-9)^2}$$

$$\text{Zeros: } f'(x) = 0 \Leftrightarrow x^4-26x^2+9=0 \Leftrightarrow (x^2)^2-26x^2+9=0 \Leftrightarrow x^2 = \frac{26 \pm \sqrt{(-26)^2-4 \cdot 9}}{2} = \frac{26 \pm \sqrt{640}}{2} \Leftrightarrow x^2 = 25,65 \vee x^2 = 0,35 \Leftrightarrow x = 5,06 \vee x = -5,06 \vee x = 0,59 \vee x = -0,59$$

$$f''(x) = \left(\frac{x^4-26x^2+9}{(x^2-9)^2} \right)' = \frac{(x^4-26x^2+9)'(x^2-9)^2 - (x^2-9)^2'(x^4-26x^2+9)}{(x^2-9)^4} = \frac{(4x^3-52x)(x^2-9)^2 - 2(x^2-9)(x^2-9)'(x^4-26x^2+9)}{(x^2-9)^4} = \frac{(4x^3-52x)(x^2-9) - 2 \cdot 2x(x^4-26x^2+9)}{(x^2-9)^3} = \frac{4x^5-88x^3+468x-4x^5+104x^3-36x}{(x^2-9)^3} = \frac{16x^3+432x}{(x^2-9)^3} = \frac{16x(x^2+27)}{(x^2-9)^3}$$

$$\text{Zeros: } f''(x)=0 \Leftrightarrow x=0$$

x		-5,06		-3		-1		-0,59		0		0,59		1		3		5.06	
f(x)	-		-	~	+	0	-		-	0	+		+	0	-	~	+		+
f'(x)	+ ↗	0	- ↘	~	- ↘		- ↘	0	+ ↗		+ ↗	0	- ↘		- ↘	~	- ↘	0	+ ↗
f''(x)	- ∩		- ∩	~	+ ∪		+ ∪		+ ∪	0	- ∩		- ∩		- ∩	~	+ ∪		+ ∪
	↗	M	↘	~	↘	Z	↘	m	↗	Z P.l.	↗	M	↘	Z	↘	~	↘	m	↗



